

Progress Check

Natural Disaster Severity Prediction

Group 5

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Problem & Dataset

Goal: predict weekly drought severity for each region over a 5–week horizon after a 91–day blind gap.

Item	Value
Regions	2,248
Training records	12,319,040
Weekly labeled targets	1,746,696
Meteorological features	14
Metric	MAE, lower is better

Key Challenges

- **Sparse labels:** `score` is only observed weekly, so most daily rows have no target.
- **Temporal mismatch:** daily weather signals must explain weekly drought severity.
- **91-day blind gap:** no ground-truth severity score is available before the test horizon.
- **Validation mismatch:** MAE values from different validation strategies are not directly comparable.

Current Implementation

The repository now supports three boosted–tree model families.

1. Build weather, seasonality, rolling, EWMA, climatology anomaly, and drought–proxy features.
2. Use 91–day–gapped historical score features to avoid touching the blind test window.
3. Train five direct horizon models: Week 1 through Week 5.
4. Blend LightGBM, XGBoost, and CatBoost outputs with `src/ensemble.py`.

Engineering status: `src/predict.py` can load LGB/XGB/CatBoost run directories.

Completed Experiments

Experiment	Validation	Local MAE	Public MAE / Status
LGBM v2	Holdout	0.6942	LGB/XGB anchor
XGBoost v1	Holdout	0.7150	LGB/XGB anchor
LGB/XGB 50/50	–	–	0.8232
CatBoost tail2737	Rolling origin	0.2212 / 0.2192 rerun	ensemble member
LGBM micro 20260520	Rolling origin	0.2002	diagnostic only
LGB/XGB/Cat 35/35/30	–	–	0.8124

Current best public record: `submissions/ensemble_20260516_lgb_xgb_cat2737_35_35_30.csv`.

Main Insight

A single local validation score does not fully explain Kaggle generalization.

- Weather-only and long-term weather models looked strong locally but were weaker on the public leaderboard.
- 91-day-gapped score history still provides important region-specific context.
- CatBoost's native categorical handling added useful ensemble diversity and improved public MAE.
- CatBoost 35%, 40%, and horizon-ramp probes did not beat the 30% public anchor.
- Rolling-origin MAE is useful, but should not be compared directly against older holdout MAE.

The next decision should be based on private robustness, not only public score.

Next Steps

- Compare the LGB/XGB anchor and CatBoost blends on the 5/22 static private leaderboard.
- Treat the 5/20 CatBoost and LGBM reruns as post-readout blend inputs, not new anchors.
- Use `docs/status/current_state.json` and `scripts/check_current_state.py` to keep report, slides, and experiment records aligned.
- Finalize the IEEE report and keep code, experiments, and submission lineage consistent.

Current status: repo-recorded best legal public MAE = **0.8124**.